

VEMIO

Inteligencia comercial con IA para CPG

Prueba Técnica — AI Product Engineer

Forecasting, Elasticidad y Uplift Promocional

Documento de Metodología y Hallazgos

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Dataset Scope: 283,533 transactions | 6 SKUs | 12 warehouses | ~25 months (Jan 2025 – Jan 2027). Zero rows discarded.

1. Data Cleaning & Business Assumptions

- **Cancelled Tickets (qty = 0, 500 rows):** Flagged with is_cancelled; excluded from demand aggregation.
- **Samples & Free Gifts (amount = 0, qty > 0, 500 rows):** Flagged with is_gift. Included in Challenge A (reflects physical inventory movement); excluded from Challenge B (zero price distorts log-log regressions).
- **Missing Metadata & Discounts (~5.5%):** Mapped using constant product_code. Discounts algebraically derived where possible ($gross = amount / (1 - discount)$). Nulls with no promo ID treated as organic sales ($discount = 0$).
- **Missing Unit Cost (product_cost, ~0.04%):** Validated as a time step-function (not constant per SKU) and imputed using monthly median cost per SKU.

2. Challenge A: Demand Forecasting

- **SKU Selection:** *Deodorant 150 ml A* (clean seasonality), *Chicken Stock 50 ct* (trend + high volume), and *Curl Shampoo 135 ml* (noisy / weak signal).
- **Validation Strategy:** Walk-forward split (last 10 of 104 weeks held out as test set; 94 training weeks). Eliminates future data leakage.
- **Primary Metric (WMAPE):** Preferred over MAPE because it weights errors by real volume sold, aligning directly with inventory replenishment risks rather than averaging unweighted percentage errors.
- **Models Evaluated:** XGBoost (lag features 1-2w, month, week of year, promo flags) vs. SARIMA (0,0,0) (without seasonal term due to 94-week history constraints).

Model Comparison & Test Performance (10-Week Test Window)

SKU	Baseline (Naive 52w)	XGBoost WMAPE	SARIMA WMAPE	Selected Model
Deodorant 150 ml A	18.9%	7.71%	58.78%	XGBoost
Chicken Stock 50 ct	10.2%	7.75%	17.32%	XGBoost
Curl Shampoo 135 ml	31.7%	14.24%	11.98%	SARIMA

Key Insight: XGBoost excels on SKUs driven by promo timing and trend. SARIMA performs better on noisy, unstructured series where machine learning models tend to overfit lag noise.

Final 8–12 Week Forecast Takeaways

- **Chicken Stock (~1,060–1,150 units):** Steady trend continuation; ready for replenishment planning.
- **Deodorant (~650–690 units, flat):** Represents a "no-promo" baseline. Requires manual uplift override when future promotional calendars are scheduled.
- **Curl Shampoo (SARIMA):** Converges to ~965 constant units after week 3; highly reliable for short-term horizon (1–3 weeks).

3. Challenge B: Price Elasticity (Deodorant 150 ml A)

- **SKU Selection:** *Deodorant 150 ml A* displayed the highest effective price Coefficient of Variation (**CV = 10.10%**, N=64,325) and strong historical demand response.
- **Econometric Model:** Log-Log Regression: $\log(quantity) = \beta_0 + \beta_1 \log(price) + \beta_2 month_dummies$. Controls for month to isolate seasonal demand spikes from price variations.

- **Result:** Elasticity $\beta_1 = -2.998$ ($p < 0.001$, $t = -19.05$, $R^2 = 0.932$). A 1% price increase drives a ~3% decrease in weekly volume.

Price Simulator & Strategic Band

Evaluated against the latest unit cost structure (~\$49.70):

- **Low Price End (\$44–\$47):** Yields negative margins (-13% to -6%) under current replacement cost. Historical sales at this price occurred when unit costs were lower (~\$45).
- **Optimal Band (\$55–\$60):** Maximizes profitability while holding healthy margins (14% to 18%), protecting volume from the severe ~62% collapse observed at the upper price ceiling (\$60.67).
- **Risks & Assumptions:** Elasticity is estimated from distributor sell-in, not consumer-level demand — forward buying around promotions likely inflates the measured sensitivity, so the true consumer-level elasticity is probably lower in magnitude. Cost is held fixed at today's level; a further cost increase would compress margins below what's projected here. Estimate is SKU-specific and should not be generalized to the rest of the catalog.

4. Challenge C: Promotional Uplift (Deodorant 150 ml A)

Compared *Combo Verano 2* (17.1% discount, 10 weeks) vs. *Combo Cierre Trimestre* (21.1% discount, 4 weeks) against a 6-week pre-promo baseline.

Metric	Summer Combo 2 (17.1% Disc.)	Q3 Close Combo (21.1% Disc.)
Volume Uplift	+7,436 units (+109.3%)	+3,574 units (+96.5%)
Margin / Unit (Normal → Promo)	\$10.40 → \$0.49	\$10.43 → -\$1.79
Incremental Units Margin	+\$3,678	-\$6,384
Base Cannibalization Margin Loss	-\$67,446	-\$45,233
Net Incremental Margin	-\$63,768	-\$51,617

- **Diagnosis:** Both promotions destroyed net margin due to heavily subsidizing base organic sales. However, *Summer 2* generated profitable incremental units (+\$3,678), whereas *Q3 Close* sold incremental volume at a direct loss (-\$1.79/unit).
- **Root Cause:** Unit costs rose ~10% over the dataset timeline without recalibrating promotional discount depths.

5. Trade-offs & AI Workflow

- **With more time or data:** consumer-level (not just distributor sell-in) transaction data would de-bias the elasticity estimate; a longer history would let SARIMA fit a full 52-week seasonal term instead of running without one; controlled discount-depth tests would replace historical uplift inference with causal evidence.
- **AI Integration:** Claude was utilized to accelerate Python code development, feature engineering, and executive prose editing. All quantitative outputs and model figures were rigorously verified against raw notebook outputs.